Bashir Agro Admin Panel
Phase 7: Reports & Analytics Module

PHASE 7 SCOPE

Implement ONLY Reports and Analytics after Phase 6.

Included:
- Dynamic reports
- Task reports
- Farmer activity reports
- Location reports
- Assignment status reports
- Activity reports
- Date filters
- Search filters
- Query optimization
- Tests

Not included:
- Flutter app
- REST API
- Task submission
- GPS validation
- Notifications

OBJECTIVE

Create a database-driven reporting system for Admin/Owner.

Reports must use real PostgreSQL data from:

- Farmers
- Farm Locations
- Task Templates
- Task Assignments
- Activity Logs

No hard-coded statistics are allowed.

ARCHITECTURE

PostgreSQL
|
Eloquent Models
|
Report Controller
|
Report Service
|
Blade Reports Page

CONTROLLERS

Create:

app/Http/Controllers/Admin/ReportController.php

Responsibilities:

- Load reports
- Receive filters
- Return report data
- Prepare export structure

SERVICE LAYER

Create:

app/Services/ReportService.php

Responsibilities:

- Calculate statistics
- Perform aggregations
- Handle date filters
- Prepare report datasets

REPORT TYPES

1. Task Performance Report

Display:

- Task Name
- Total Assignments
- Completed
- In Progress
- Cancelled
- Completion Percentage

Source:
task_templates
task_assignments

2. Farmer Activity Report

Display:

- Farmer Name
- Total Assigned Tasks
- Completed Tasks
- Pending Tasks

- Completion Rate

Source:
farmers
task_assignments

3. Location Performance Report

Display:

- Location Name
- Assigned Tasks
- Completed Tasks
- Active Farmers

Source:
farm_locations
task_assignments
farmer_location

4. Assignment Status Report

Display:

- Assigned
- In Progress
- Completed
- Cancelled

Filters:
- Date
- Priority
- Location

5. System Activity Report

Source:

activity_logs

Display:

- User
- Action
- Module
- Description
- Date
- Time

DATE FILTERS

Support:

- Today
- Yesterday
- This Week
- This Month
- Custom Date Range

Example:

start_date=2026-10-01
end_date=2026-10-31

SEARCH AND FILTERS

Task reports:
- Task name
- Category

Farmer reports:
- Farmer name
- Activity

Location reports:
- Location name

Assignment reports:
- Status
- Priority

UI REQUIREMENTS

Preserve Bashir Agro design:

- Bengali UI
- Green theme
- Existing cards
- Tables
- Responsive layout

Create:

resources/views/reports/index.blade.php

DATABASE RULES

Use:

- Eloquent relationships
- Aggregation queries
- Count()
- GroupBy()
- Date filtering

Avoid:

- Loading unnecessary records
- N+1 queries
- Complex logic inside Blade

PERFORMANCE

Implement:

- Eager loading
- Database indexes
- Pagination
- Optimized queries

CHARTS

Optional visualization:

- Completion percentage
- Status distribution
- Farmer workload

Charts are only presentation.
Data must come from Laravel.

ROUTES

Create:

GET /admin/reports

Optional:

GET /admin/reports/task
GET /admin/reports/farmers
GET /admin/reports/locations

SECURITY

Use:

- Authentication middleware
- Authorization
- Validated filters
- Safe query building

TESTS

Create:

tests/Feature/ReportsTest.php

Test:

- Admin access
- Guest restriction
- Date filters
- Search filters
- Task calculations
- Farmer calculations
- Location calculations
- Activity report loading

ANTIGRAVITY INSTRUCTION

Proceed to PHASE 7 ONLY: Reports and Analytics Module.

Use the existing Bashir Agro Laravel project after Phase 6.

Do not redesign the UI.

Do not implement:

- Flutter app
- REST API
- Mobile reports
- Task submission
- GPS validation
- Notifications

Implement:

- ReportController
- ReportService
- Task performance reports
- Farmer activity reports
- Location reports
- Assignment status reports
- Activity reports
- Date filters
- Search filters
- Query optimization
- Feature tests

Preserve:

- Bengali UI
- Green agricultural theme
- Existing cards
- Existing tables
- Responsive design

Database must be the source of truth.

Remove all hard-coded report values.

After implementation:

Run:

php artisan test

Run:

vendor/bin/pint

Report:

- Files created
- Files modified
- Routes added
- Database changes
- Test results

STOP after Phase 7.

Do not start Phase 8 automatically.